

LESSON 2: PREDICTION

In the previous lesson, we explored how an AI system learns using machine learning. It analyzes data, identifies patterns, and builds a model that helps it understand and make sense of new inputs. For example, after studying many examples, the AI forms an internal model of what a 'cup' looks like. This model becomes the basis for making predictions when it encounters a new object.

In this lesson, we will explore the basics of prediction and how Big Data helps AI make more accurate and reliable predictions.

PREDICTION

The answer to something is not always obvious. In many situations, the outcome isn't clear, and we have to make a guess. When we do, we rely on our experience and knowledge to make the best guess we can — even if we're not completely sure.

Prediction is the process by which an intelligent system — including humans — tries to estimate the most likely answer or outcome in a new situation.

PREDICTION IS WHEN AN AI USES ITS MODEL TO ESTIMATE THE MOST PROBABLE OUTPUT.

EXAMPLE 2.1

When Maya first learned to associate the object <cup> with the word 'cup', her mental model was very simple — it was based on just one cup. At first, she guessed correctly when she saw similar-looking cups. But when she saw cups that looked different — maybe taller, wider, or with a lid — she sometimes guessed wrong.

Each time she made a wrong prediction, her parents corrected her. Maya learned from her mistake and updated her model of what a cup could look like.

Over time, with more examples and feedback, her predictions improved. She became better at correctly classifying new objects as cup or not a cup — even if they looked different from the first one she saw.



Maya is learning to recognize cups. With each new example, her ability to recognize cups improves — increasing the probability that she will predict correctly, even when they look different.

In practice, prediction in AI takes different forms depending on the type of data it works with. In general, prediction applies to two kinds of data: categorical and continuous. In both cases, the AI draws on patterns it has learned from previous data to make its best guess — a prediction that reflects the most probable outcome given the information it has.

Whether it is predicting a category or estimating a numerical value, the model is performing the same essential task: using what it has learned to make a well-informed prediction in a new situation.

CATEGORICAL PREDICTION

Sometimes, the goal is to decide which category something belongs to. For example, the AI might try to recognize whether an object is a cat, determine the language of a spoken word, or match an animal sound to its species. These are all examples of categorical prediction, where the AI chooses the most likely label from a set of possible categories — much like answering the question, “What is this?”

EXAMPLE 2.2

At a recycling facility, a robotic system separates plastic bottles from other waste. A camera captures images of objects on a moving conveyor belt and sends them to a prediction model trained to recognize plastic bottles. The model analyzes each image and estimates the probability that the object is a plastic bottle. If the probability is high enough, the system triggers a robotic arm to remove the object and place it in the plastic recycling bin.



An AI-controlled robotic arm identifies and removes a plastic bottle from a moving conveyor belt at a recycling facility.

When the system began operating, it was tested with 100 items — 50 plastic bottles and 50 non-bottle items. During the test, the AI model made 30 mistakes.

- It failed to identify 20 plastic bottles, meaning it correctly classified only 30 out of 50 bottles.
- It mistakenly classified 10 non-bottle items as plastic bottles, meaning it correctly identified 40 non-bottle items as not bottles.

In total, the system got 70 out of 100 predictions correct, resulting in an accuracy of 70%.

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} = \frac{30 + 40}{100} = \frac{70}{100} = 70\%$$

The items that were misclassified were collected and reviewed to identify the reasons for these errors, helping the team understand how the model could be improved.

- **Bottles Missed:** The model failed to recognize green bottles and crumpled bottles, which looked different from the training examples.
- **Objects Misclassified:** The model incorrectly classified these objects as plastic bottles due to similar shapes or features.



Original Data

Bottles it was trained on.



Bottles Missed

The model missed green bottles and crumpled bottles.



Objects Misclassified

The model misclassified objects that had similar shapes to bottles it was trained on.

ACCURACY IS ONE MEASURE OF A MODEL'S PERFORMANCE.

To improve the model's predictive accuracy and reduce its errors, the AI was retrained using a larger and more diverse set of images. This new training data included:

- Bottles in various colors (green, blue, and clear)
- Bottles that were crushed, dented, or partially obscured
- Images taken from different angles and lighting conditions
- More examples of non-bottle items with similar shapes or materials

By increasing the volume and variety of training data, the model learned to recognize a wider range of bottles and distinguish them more reliably from other objects.

The updated model was tested again using a new set of 100 items, with the same distribution as before: 50 plastic bottles and 50 non-bottles. This time, the model made only 10 mistakes:

- It missed 5 actual plastic bottles,
- And it misclassified 5 non-bottles as bottles.

That means it correctly identified 45 plastic bottles and 45 non-bottle items, giving a new accuracy of:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} = \frac{45 + 45}{100} = \frac{90}{100} = 90\%$$

GREATER VOLUME AND VARIETY OF DATA IMPROVE A MODEL'S PREDICTIVE ACCURACY.

CONTINUOUS PREDICTION

In the previous section, we explored categorical prediction, which applies to categorical data and involves assigning inputs to predefined categories. But in many cases, the task is not to classify, but to estimate a number. The AI might be asked to predict tomorrow's temperature, estimate how many points a student will score on a test, or forecast how much it will rain.

These are all examples of continuous prediction, where the goal is not to choose a category but to estimate a numerical value — answering questions like “How much?” or “What value?”

EXAMPLE 2.3

In a fictional neighborhood, an AI system is being developed to help estimate house prices based on their size. The only information the AI receives is the floor area of each house (in square meters) and the price it sold for. The goal is to train the AI to recognize the pattern between size and price so that it can predict the price of a new house based only on its size.

To begin, the AI is trained on a small dataset containing the sizes and sale prices of just six houses. Using this limited information, the AI attempts to learn the relationship between a house's size and its price by identifying a pattern in the data.



Size (square meters)	Price (\$1,000's)
107	291
115	292
107	302
115	305
93	227
82	173
100	274
120	340
130	316
90	234

This figure shows the data used to teach the AI model how house size relates to price. The scatter plot displays the actual data points, while the red dashed line represents the best-fit line created by the model. The table lists the size and sale price of the ten houses used to train the model.

The model derives the equation for the best-fit line and uses it to predict prices based on house size. This equation allows the AI to estimate the selling price of any house in the neighborhood, based solely on its size. In this case, the equation is:

$$\text{Price} = 2.53 \times \text{Size} + 18.45$$

Now the model must be tested. It is given the sizes of five houses it has not encountered before. Using its best-fit line equation, the model predicts the price of each house based solely on its size.

To determine how well the prediction model performs, the predicted prices are compared to the actual sales prices. The difference between each predicted and actual value is called the error.

$$\text{Error} = \text{Actual Price} - \text{Predicted Price}$$

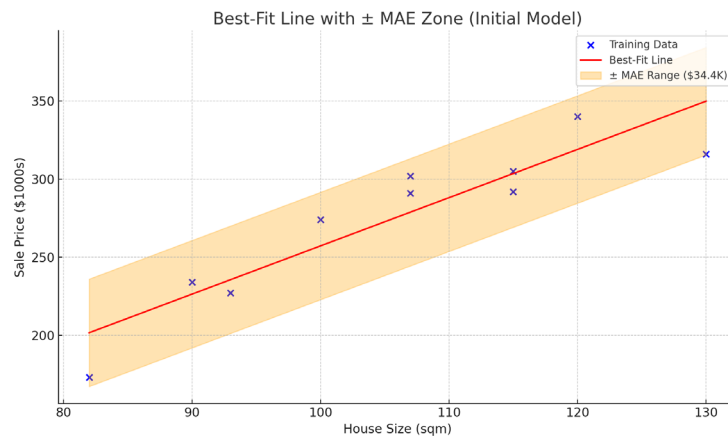
Size (sqm)	Actual Price (\$1,000)	Predicted Price (\$1,000)	Error (\$1,000)
111	291	124	-52
124	331	316	-15
157	433	406	-27
114	300	264	-36
146	399	357	-42

This table shows the actual and predicted prices for five houses that were not part of the training data. The error is calculated by subtracting the predicted price from the actual price. All values are negative, indicating that the model consistently underestimated the true prices.

A positive error means the model underestimated the price, while a negative error means it overestimated. To measure overall performance, we calculate the Mean Absolute Error (MAE) — the average size of the errors, using their absolute values to ignore direction. This tells us how far off the predictions are, on average.

$$\text{MAE} = \frac{\text{Sum of absolute value of errors}}{\text{Total Predictions}} = \frac{52 + 15 + 27 + 36 + 42}{5} = 34.4$$

The best-fit line shows what the model predicts, but the real-world value won't always land exactly on that line. The MAE gives us a typical margin of error, so we expect actual prices to fall within a band around the line — about plus or minus the MAE.



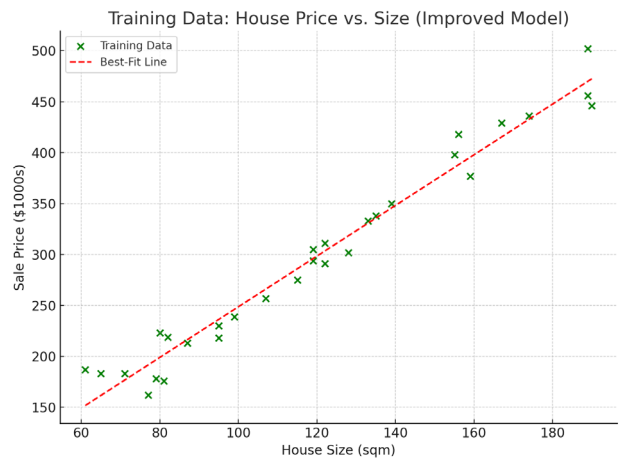
This graph shows the best-fit line produced by the model. The orange shaded area represents the full Mean Absolute Error (MAE) of \$34,400, illustrating the average range within which actual prices may fall around each predicted value.

THE MEAN ABSOLUTE ERROR IS ONE MEASURE OF A MODEL'S PERFORMANCE.

To improve the predictive performance of the model, it is trained on a larger volume and more varied set of data. Not only was it trained on a greater number of houses, but also on a wider range of sizes and prices. In the initial dataset, house sizes ranged from 82 to 130 square meters. Now, the model is exposed to houses both smaller and larger, with sizes ranging from 61 to 190 square meters.

The model recalculates the best-fit line equation using the expanded dataset and applies it to generate new predictions.

$$\text{Price} = 2.49 \times \text{Size} + 0.14$$



This graph shows the expanded training dataset, consisting of 30 houses with sizes ranging from 61 to 190 square meters. Each green point represents one house, and the red dashed line is the best-fit line generated by the model to represent the relationship between house size and price.

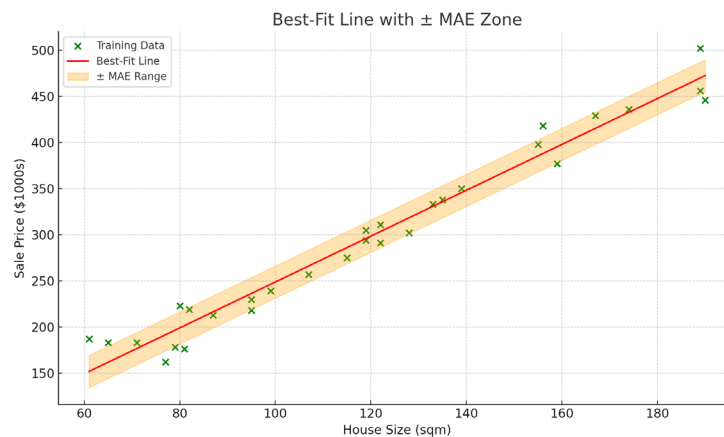
Using the new best-fit line, the model is tested again on the same five house sizes as before. For each, it predicts a price, which is then compared to the actual sale price. This allows us to calculate a new MAE, and evaluate whether the model’s performance has improved.

Size (sqm)	Actual Price (\$1,000)	Predicted Price (\$1,000)	Error (\$1,000)
111	239	276	-37
124	316	308	8
157	406	390	16
114	264	284	-20
146	357	363	-6

This table shows the model’s predictions after retraining on a larger dataset. For each house, the predicted price is compared to the actual sale price, and the error is calculated. Positive errors indicate underestimation; negative errors indicate overestimation.

$$MAE = \frac{\text{Sum of absolute value of errors}}{\text{Total Predictions}} = \frac{37 + 8 + 16 + 20 + 6}{5} = 17.4$$

As the volume and variety of training data increase, the MAE decreases. This means the zone of error around the best-fit line becomes narrower, making the model's predictions closer to the actual values.



This graph shows the best-fit line generated by the improved model, trained on a larger and more varied dataset. The orange shaded area represents the full Mean Absolute Error (MAE) of \$17,400 — a much narrower range than in the previous model, which had an MAE of \$34,400. The reduced error band indicates that the model's predictions are now, on average, closer to the actual sale prices.

Notes

Common Misconceptions

- An error means a bad prediction: All models make errors. The goal is to minimize them, not eliminate them — errors help the model improve.
- More data means a perfect model: More data helps, but only when it includes useful variety. Repetition or poor-quality data can make things worse.
- Models only learn from correct predictions: Models learn from all outcomes — including incorrect predictions — by using errors to adjust and improve over time.

CONCLUSION

In this lesson, we explored how AI makes predictions using two types of data: categorical and continuous. For categorical data, we measure how well the model performs using accuracy — the percentage of correct predictions. For continuous data, we use Mean Absolute Error (MAE) to calculate how far off the predictions are, on average.

We also saw that an AI model's ability to make accurate predictions depends on the data it learns from. As we increased the volume of training data and introduced greater variety, the model's predictions became more reliable. These two ideas — volume and variety — are part of what we call Big Data, which plays a key role in improving AI performance.

SUMMARY

Prediction is the process of estimating an outcome based on patterns learned from data.

AI can make predictions using:

- **Categorical** data (e.g., identifying whether an object is a bottle)
- **Continuous** data (e.g., estimating house prices or rainfall)

Accuracy is used to evaluate predictions on categorical data by measuring how many were correct out of all attempts.

Mean Absolute Error (MAE) is used for continuous data to measure how far off the predictions are, on average.

All models make errors — the goal is to reduce them, not eliminate them.

AI models improve as they are trained on more data. Specifically:

- **Volume** (more data) helps the model learn broader patterns.
- **Variety** (more types of examples) helps the model generalize better.

These concepts are part of what is called **Big Data**, which supports better learning and more accurate predictions in AI systems.

Activity 2.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	AI models can make predictions using either categorical or continuous data.
<u>F</u>	Accuracy is the best way to measure how close a numerical prediction is to the actual value.
<u>T</u>	Mean Absolute Error (MAE) tells us the average distance between predicted and actual values.
<u>F</u>	A model that makes prediction errors is considered broken.
<u>F</u>	More data always makes a model perfect.
<u>T</u>	Volume refers to the total amount of training data available to the model.
<u>F</u>	Variety means training a model on examples that are all the same type.
<u>T</u>	A prediction model for continuous data will produce a number, not a label.
<u>F</u>	If a model has high accuracy, that means its MAE is also low.
<u>T</u>	Big Data helps improve AI performance by increasing both volume and variety of training data.

Activity 2.2

Answer the questions below.

1. What is the goal of prediction in AI?
 - a) To memorize all data exactly
 - b) To estimate the most likely outcome using learned patterns
 - c) To randomly guess based on probability
 - d) To classify only images
2. Which of the following is an example of a categorical prediction?
 - a) Estimating the temperature tomorrow
 - b) Predicting a student's exam score
 - c) Determining if an object is a plastic bottle
 - d) Calculating total sales for a store
3. Which metric can be used to evaluate continuous predictions like temperature?
 - a) Accuracy
 - b) Precision
 - c) Recall
 - d) Mean Absolute Error (MAE)
4. What does a high MAE value tell us about a model's predictions?
 - a) The model never makes mistakes
 - b) The model's predictions are far from actual values
 - c) The model's predictions are always correct
 - d) The model is overfitting
5. How is MAE calculated?
 - a) Percentage of correct predictions
 - b) Average of the absolute differences between predicted and actual values
 - c) Number of categories predicted correctly
 - d) The square root of total errors

Activity 2.2 (continued)

Answer the questions below.

6. What does the term "volume" refer to in the context of training data?
 - a) The total amount of data the model learns from
 - b) The range of model outputs
 - c) The number of times a model makes errors
 - d) The speed at which the data is processed
7. Why is "variety" in training data important for improving a model?
 - a) It increases the speed of training
 - b) It reduces the cost of data collection
 - c) It prevents the use of continuous prediction
 - d) It helps the model generalize to different examples
8. Which best describes the difference between accuracy and MAE?
 - a) MAE is used for text predictions; accuracy is used for numbers
 - b) Both are used for categorical predictions
 - c) Accuracy is used for categorical data; MAE is used for continuous data
 - d) Accuracy and MAE mean the same thing
9. What happens when a model is trained on a small, narrow dataset?
 - a) It performs well on all tasks
 - b) It automatically adjusts its error
 - c) It always predicts perfectly
 - d) It struggles to make accurate predictions on new data
10. Which of the following statements is true?
 - a) Models only learn from correct predictions
 - b) Models learn from both correct and incorrect predictions
 - c) Models stop learning after one training cycle
 - d) Models only learn from testing data

Activity 2.3

Read the narrative and answer the questions below.

A software company is developing an AI system that can automatically detect the language of short text messages. To train the model, engineers collect thousands of short sentences labeled with the language they are written in — such as English, Spanish, Arabic, and French.

The model learns to recognize specific features like vocabulary, accents, and sentence structure. Once trained, it is tested on new sentences submitted by users. It correctly identifies most languages, but it struggles with shorter messages or those that use informal spelling.

After reviewing the errors, the engineers realize that the AI hadn't been trained on enough casual or abbreviated text. They improve the model by adding more examples that include informal language, slang, and abbreviations. The system becomes more accurate and consistent in real-world use.

1. What type of learning was used to train the AI system to recognize languages, and how do you know?

Supervised learning — because the model was trained using labeled examples, where each sentence was tagged with its language.

2. What kind of prediction is this AI system performing?

Categorical prediction — the model is classifying each sentence into one of several languages.

3. Why did the AI model struggle with some user-submitted sentences?

Because the model hadn't been trained on enough informal or abbreviated text, so it couldn't recognize the language accurately in those cases.

4. How did the engineers improve the model?

They retrained it using more varied examples, including informal language and slang.

5. Why is it important to train the model with a wide variety of examples?

So the model can generalize and make accurate predictions, even when input appears in unfamiliar or casual forms.

Activity 2.4

Read the narrative and answer the questions below.

A transportation company is building an AI system that can estimate how long a trip will take by car. To train the model, engineers collect thousands of examples showing how long it took to travel different distances at different times of day. Each example includes the distance traveled and the actual travel time in minutes.

The AI is first trained on a small set of data — mostly from trips taken during off-peak hours. From this, it learns a basic pattern: longer distances usually mean longer travel times. But when tested on trips during rush hour, the model often predicts travel times that are too short, because it hasn't learned how traffic affects the results.

To fix this, the engineers expand the training data to include trips taken at different times of day, including busy periods with heavy traffic. The model adjusts its understanding and begins to make more accurate predictions.

1. What kind of prediction is the AI model making in this scenario?

Continuous prediction — it's estimating a numerical value (travel time in minutes).

2. Why were the model's early predictions inaccurate during rush hour?

Because the model had not been trained on examples that included traffic delays.

3. How did the engineers improve the model's performance?

They added more varied training data from different times of day.

4. What is one way to measure how far off the model's predictions are?

By calculating the Mean Absolute Error (MAE), which gives the average difference between predicted and actual travel times.

5. Why is it important to include both volume and variety in training data?

So the model can make accurate predictions across different conditions, such as heavy or light traffic.

Activity 2.5

Read the scenario below. Then answer the questions that follow. Use the provided spreadsheet. You will practice sorting, filtering, and using formulas—just like real data scientists do when they test AI models. Use the short link to access the spreadsheet: bit.ly/C4_AI_Activity2_5

- a. A grocery distribution center is testing an AI model that sorts fruit. The model decides if each item is an Apple or Not Apple. The center tests the model on a batch of 50 fruits. For each item, the AI makes a prediction. The spreadsheet shows both the actual label and the AI prediction.

1. How many predictions were correct?	36
2. How many predictions were incorrect?	14
3. What was the model's overall accuracy?	72%
4. How many apples were missed?	7
5. How many tomatoes were misclassified?	11

6. Which fruit did the AI confuse with apples? Why do you think this happened?

Tomatoes — confused with apples because they are round, similar in size, and often red like apples

- b. The grocery distribution center is considering using the AI model for full operations. On average, the center handles 100,000 apples every week, along with other fruits. When the AI makes a mistake, it costs the business money:

- Missing an apple = \$5 lost
- Mistaking a tomato for an apple = \$1 wasted

1. How many apples would be missed every week?	14,000
2. How many tomatoes would be misclassified every week?	22,000
3. What would the total weekly cost of missed apples be?	\$70,000
4. What would the total weekly cost of misclassified tomatoes be?	\$22,000
5. What is the combined weekly cost of using this AI model?	\$92,000

6. Based on these numbers, do you think the AI is ready for real-world use? Why?

The AI is not ready for operations — the cost of errors is extremely high. The model would need more training or a different approach before being deployed.

Activity 2.5 (continued)

Read the scenario below. Then answer the questions that follow. Use the provided spreadsheet. You will practice sorting, filtering, and using formulas—just like real data scientists do when they test AI models. Use the short link to access the spreadsheet: bit.ly/C4_AI_Activity2_5

- c. After reviewing the high cost of errors, the distribution center retrained its AI model to better distinguish between apples and tomatoes.

Refer to the sheet 'Activity 2.5c' to answer the following questions.

1. What is the accuracy of the new model?	90%
2. How many apples would it miss out of 100,000?	4,000
3. How many tomatoes would it misclassify out of 100,000?	6,000
4. What would the weekly cost of missed apples be?	\$20,000
5. What would the total weekly cost of misclassified tomatoes be?	\$6,000
6. What is the total weekly cost of errors with the new model?	\$26,000

7. How does this compare to the old model in Activity 2.5b? What is the improvement?

Old model cost = \$92,000 per week. New model cost = \$26,000 per week.

Savings = \$66,000 per week.

8. Think about another area of life. Where would AI errors be more serious than in a grocery store?

AI errors would be more serious in areas where lives are at stake. Examples:

Healthcare: A wrong diagnosis could harm a patient.

Traffic/Transportation: A self-driving car could cause an accident.

Airlines: An AI error in navigation or maintenance could risk hundreds of lives.

9. Imagine you are in charge of the distribution center. Would you use the 90% accurate AI, or keep humans sorting apples, or a combination of both AI and human? Explain your choice.

I would choose a combination of AI and human workers. The AI is fast and saves money, but at 90% accuracy it still makes mistakes. By adding human quality control, people can catch the errors the AI misses. In the real world, this is called 'Human-in-the-Loop AI'. AI and humans working together are more effective than either on their own.

Activity 2.6

Read the scenario below, then answer the questions that follow. Use the provided spreadsheet to apply formulas and calculations—just like real data scientists do when they test AI models. Use the short link to access the spreadsheet: bit.ly/C4_AI_Activity2_6

A company is testing an AI model to predict how many units of a product it will sell at different prices. Accurate predictions help the company plan inventory and revenue.

The AI studied data and built the following best-fit line:

$$\text{Sales (units)} = -12 \times \text{Price (\$)} + 220$$

The company then tested the model at different prices between \$8.00 and \$12.50.

- Complete the table by using the equation of the best-fit line to predict the number of units sold and the error for each prediction.

Price (\$)	Actual Units Sold (units)	Predicted Units Sold (units)	Error (units)	Error (Absolute value)
8.0	115	124	-9	9
9.0	125	112	13	13
10.0	95	100	-5	5
11.0	101	88	13	13
12.0	67	76	-9	9

- Calculate the Mean Absolute Error.

The MAE = 9.80

- If the AI makes one prediction every day, then over a 30-day month, by how many units will the company misestimate sales?

Total Error (units) = MAE x Days = 9.80 x 30 = 294 units

- If the company sells 10,000 units in a month, what percentage of sales does the total error (294 units) represent? What does this tell us about the model's usefulness?

294/10,000 x 100 = 2.94%

The model's error is about 3% of total sales. It is fairly accurate, but lowering the error could still save hundreds of units each month.

